SHEET 4

SIC/XE Machine

Example 1:

Translate (by hand) the following assembly program to SIC object code. Which contains H record, T record, and E record.

```
SUM       Start   0000
          LDX     #0
          LDA     #0
          +LDB    #TABLE2
          BASE    TABLE2
LOOP      ADD     TABLE,X
          ADD     TABLE2,X
          TIX     COUNT
          JLT     LOOP
          +STA    TOTAL
          RSUB
COUNT     RESW    1
TABLE     RESW    2000
TABLE2    RESW    2000
TOTAL     RESW    1
          END     SUM
```

Example 2:

Translate (by hand) the following assembly program to SIC object code. Which contains H record, T record, and E record.

00000	COPY	Start	0000
000		LDS	#3
0003		LDT	#300
0006		LDX	#0
0009		LDA	ALPHA,X
000C	ADDLP	ADD	BETA,X
000F		STA	GAMMA,X
0012		ADDR	S,X
0015		COMPR	X,T
0017		JLT	ADDLP
0019		RESW	100
001C	ALPHA	RESW	100
0148	BETA	RESW	100
0274	GAMMA	END	COPY
03A0			

Symbols	Address
ADDLP	000C
ALPHA	001C
BETA	0148
GAMMA	03A0

Example 3:

Translate (by hand) the following assembly program to SIC object code. Which contains H record, T record, and E record.

```
COPY      Start     0000
FIRST     STL       RETADR
          LDB       #LENGTH
          BASE      LENGTH
CLOOP     +JSUB     RDREC
          LDA       LENGTH
          COMP      #0
          JEQ       ENDFIL
          +JSUB     WRREC
          J         CLOOP
ENDFIL    LDA       EOF
          STA       BUFFER
          LDA       #3
          STA       LENGTH
          +JSUB     WRREC
          J         @RETADR
EOF       BYTE      C'EOF'
RETADR    RESW      1
LENGTH    RESW      1
BUFFER    RESB      4096
RDREC     CLEAR     X
          CLEAR     A
          CLEAR     S
          +LDT      #4096
RLOOP     TD        INPUT
          JEQ       RLOOP
          RD        INPUT
          COMPR     A,S
          JEQ       EXIT
          STCH      BUFFER,X
          TIXR      T
          JLT       RLOOP
EXIT      STX       LENGTH
```